

Differential Equations Lesson 11

Question 1:

Solve the differential equation $y'' - 9y = 6x - 9x^3$ using the particular solution $y_p = x^3$.

Note: $L(y) = y'' - 9y = 6x - 9x^3$; and the complementary equation $L(y) = 0$ is $y'' - 9y = 0$.

a) The general solution of $y'' - 4y = 0$ is $y_c = \underline{\hspace{2cm}}?$

b) The general solution of $L(y) = y'' - 9y = 6x - 9x^3$ is $y = y_p + y_c = \underline{\hspace{2cm}}?$

Question 2:

Solve the differential equation $y'' - 9y = x - 9x^2$ using the particular solution $y_p = x^2 - \frac{x}{9} + \frac{2}{9}$.

Note: $L(y) = y'' - 9y = x - 9x^2$; and the complementary equation $L(y) = 0$ is $y'' - 9y = 0$.

a) The general solution of $y'' - 9y = 0$ is $y_c = \underline{\hspace{2cm}}?$

b) The general solution of $L(y) = y'' - 9y = x - 9x^2$ is $y = y_p + y_c = \underline{\hspace{2cm}}?$

Question 3:

Solve the differential equation $y'' + 8y' - 9y = 4x$ using the particular solution $y_p = -\frac{4x}{9} - \frac{32}{81}$.

Note: $L(y) = y'' + 8y' - 9y = 4x$; and the complementary equation $L(y) = 0$ is $y'' + 8y' - 9y = 0$.

a) The general solution of $y'' + 8y' - 9y = 0$ is $y_c = \underline{\hspace{2cm}}?$

b) The general solution of $L(y) = y'' + 8y' - 9y = 4x$ is $y = y_p + y_c = \underline{\hspace{2cm}}?$

Question 4:

Solve the differential equation $y'' + 4y' - 5y = 4x^2$ using the particular solution $y_p = -\frac{4x^2}{5} - \frac{32x}{25} - \frac{168}{125}$.

Note: $L(y) = y'' + 4y' - 5y = 4x^2$; and the complementary equation $L(y) = 0$ is $y'' + 4y' - 5y = 0$.

a) The general solution of $y'' + 4y' - 5y = 0$ is $y_c = \underline{\hspace{2cm}}?$

b) The general solution of $L(y) = y'' + 4y' - 5y = 4x^2$ is $y = y_p + y_c = \underline{\hspace{2cm}}?$

Question 5:

Solve the differential equation $y'' + 3y' - 4y = x - 2x^2$. The particular solution y_p is NOT given.

a) Finding the general solution of the complementary equation $L(y) = y'' + 3y' - 4y = 0$.

Hint: The characteristic equation is $r^2 + 3r - 4 = 0$ or $(r + 4)(r - 1) = 0$; and the roots are $r = -4$ and 1 .

The general solution of $L(y) = y'' + 3y' - 4y = 0$ is $y_c = \underline{\hspace{2cm}}?$.

b) Find a particular solution (y_p) of the differential equation $y'' + 3y' - 4y = x - 2x^2$. $y_p = \underline{\hspace{2cm}}?$

c) The general solution of the differential equation $y'' + 3y' - 4y = x - 2x^2$ is $y_p + y_c = \underline{\hspace{2cm}}?$.

Question 6:

Solve the differential equation $y'' - 8y' + 25y = x^2 + 3x$. The particular solution y_p is NOT given.

a) Find the general solution of the complementary equation $L(y) = y'' - 8y' + 25y = 0$.

The characteristic equation is $r^2 - 8r + 25 = 0$; and the roots are $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{8 \pm \sqrt{-36}}{2} = 4 \pm 3i$.

The general solution of the the complementary equation $L(y) = y'' - 8y' + 25y = 0$. $y_c = \underline{\hspace{2cm}}?$

b) Find a particular solution (y_p) of the differential equation $y'' - 8y' + 25y = x^2 + 3x$. $y_p = \underline{\hspace{2cm}}?$

c) Find the general solution of the differential equation $y'' - 8y' + 25y = x^2 + 3x$.

$y = y_c + y_p = \underline{\hspace{2cm}}?$

Question 7:

Solve the differential equation $y'' + 4y = \sin x$. The particular solution y_p is NOT given.

- a) The complementary equation is $L(y) = y'' + 4y = 0$.

Hint: The characteristic equation is $r^2 + 4 = 0$; and the roots are $r = \pm 2i = 0 \pm 2i$.

The general solution of the complementary equation $L(y) = y'' + 4y = 0$ is $y_c = \underline{\hspace{2cm}}?$

- b) Find a particular solution (y_p) of $y'' + 4y = \sin x$.

Hint: Let $y_p = x(A \cos x + B \sin x)$

$y_p = \underline{\hspace{2cm}}?$

- c) The general solution of the differential equation $y'' + 4y = \sin x$ is

$y = y_c + y_p = \underline{\hspace{2cm}}?$

Question 8:

Solve the differential equation $y'' + 8y' + 16y = e^{3x}$. The particular solution y_p is NOT given.

a) The complementary equation is $L(y) = y'' + 8y' + 16y = 0$.

Hint: The characteristic equation is $r^2 + 8r + 16 = 0$ or $(r + 4)(r + 4) = 0$; and the repeated real root is $r = -4$.

The general solution of the complementary equation $L(y) = y'' + 8y' + 16y = 0$ is $y_c = \underline{\hspace{2cm}}?$

b) Find a particular solution (y_p) of $y'' + 8y' + 16y = e^{3x}$.

Hint: Let $y_p = A \cdot e^{3x}$. Find $(y_p)'$ and $(y_p)''$.

Substitute y_p , $(y_p)'$ and $(y_p)''$ into $y'' + 8y' + 16y = e^{3x}$ to find A .

$y_p = \underline{\hspace{2cm}}?$

c) The general solution of the differential equation $y'' + 8y' + 16y = e^{3x}$ is

$y = y_c + y_p = \underline{\hspace{2cm}}?$

